

Data Cleaning & Preparation — Change Log

DecodeLabs | Data Analytics Track | Project 1 | Batch 2026

Dataset Overview

Source file: **Dataset_for_Data_Analytics.xlsx** — an e-commerce order transactions export containing 1,200 rows and 14 columns (OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice).

Summary of Data Quality Issues Found

Issue	Rows Affected	Severity
Missing CouponCode values	309 / 1,200 (25.75%)	Low — valid business scenario
Exact duplicate rows	0	None found
Duplicate OrderID (unique identifier)	0	None found
Duplicate TrackingNumber	0	None found
TotalPrice inconsistent with Qty x UnitPrice	0	None found
Statistical outliers in TotalPrice (IQR method)	8 / 1,200 (0.67%)	Low — flagged, not removed

Change Log — What Was Changed and Why

Change ID	Description	Impact	Status
CR001	Imputed missing CouponCode with label 'NoCoupon'	Preserved all 309 orders; no rows dropped	Resolved
CR002	Verified & confirmed zero exact duplicate rows	No rows removed (dataset already clean)	Verified
CR003	Verified & confirmed zero duplicate OrderID / TrackingNumber	Confirms referential integrity of 1,200 unique orders	Verified
CR004	Trimmed whitespace & standardised case on 6 text columns	Improved consistency for grouping/filtering	Resolved
CR005	Enforced ISO 8601 date format (YYYY-MM-DD) on Date column	0% date format errors confirmed	Resolved
CR006	Rounded UnitPrice & TotalPrice to 2 decimal places	Consistent currency precision	Resolved

CR007	Cross-validated TotalPrice against Quantity x UnitPrice	0 mismatches found; no correction needed	Verified
CR008	Flagged 8 statistical outliers in TotalPrice (IQR method)	Retained as valid bulk orders; added IsPriceOutlier flag column	Resolved
CR009	Corrected column data types (dates, categories, IDs)	Enables reliable downstream analysis	Resolved

Verification Gate — Threshold for Project 2

Requirement	Result
0% error rate on unique identifiers (OrderID)	PASS — 0 duplicates
0% error rate on date formats	PASS — 100% ISO 8601 compliant
0 remaining null values	PASS — all nulls resolved

Tools & Methodology

Cleaning was performed using **Python (pandas)** in a Jupyter Notebook (*Data_Cleaning_Preparation.ipynb*), chosen for reproducibility and scalability beyond what a manual Excel/Power Query workflow allows. All transformation steps are scripted and re-runnable against future data exports.

Prepared by: Data Analytics Intern | **Reviewed against:** DecodeLabs Project 1 Verification Gate | **Output:** cleaned_dataset.xlsx / .csv